Generalized Morse theory for distance functions

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1 Generalities on Lipschitz Functions

1.1 Lipschitz functions on manifolds and subdifferentials

1.1.1 General facts and definitions

For all the following, M will denote a differentiable manifold of dimension m. In this section, we generalize many definitions related to Lipschitz functions. We begin by extending the definition of Lipschitz functions to functions on a manifold.

Definition 1. *A function $f : M \to \mathbb{R}^n$, where M is a smooth manifold, is locally Lipschitz if for any chart $U \subset M$ and any compact $K \subset U$, there exists $C > 0$ such that, with the coordinates on U*

$$\forall x, y \in K, |f(x) - f(y)| \leq C|x - y|$$

Let $f : U \subset \mathbb{R}^m \to \mathbb{R}^n$ be a Lipschitz function, where U is open. Rademacher's theorem, which states that f is differentiable almost everywhere, will allow us to generalize differentials to this setting.

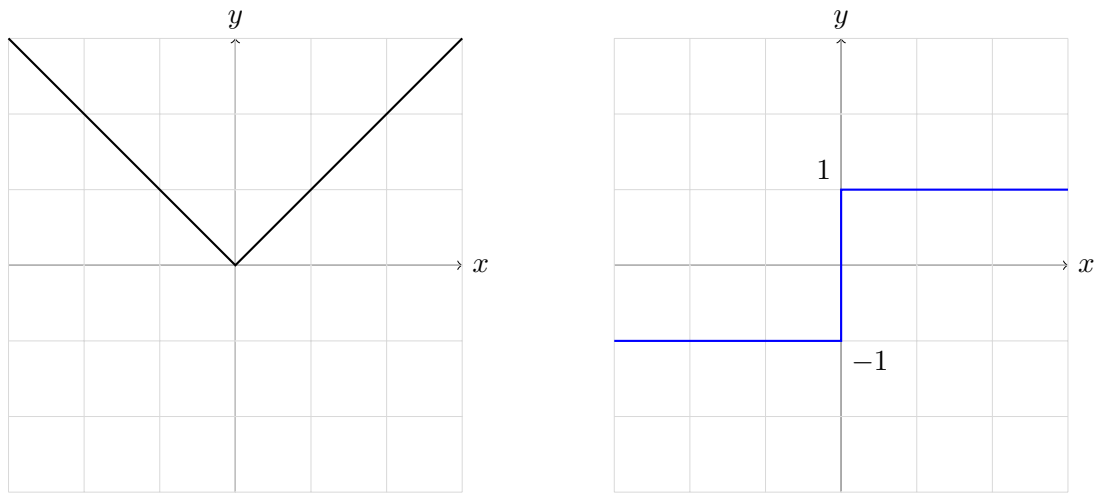
Definition 2. *[6] The subdifferential of f at x, denoted by $\partial_x f$ is the convex hull of the linear maps l obtained as*

$$l = \lim_{i \to \infty} d_{x_i} f$$

where f is differentiable at each x_i and $x_i \to x$.

The subdifferential is then set valued. This set is always non empty and compact, f being Lipschitz bounds df where it is defined.
By [5], we can exclude a set of measure 0 from the possible values of x_i without changing the definition.

Figure 1: The absolute value and its subdifferential represented in the plane
The set on the right is $\bigcup_{l \in \partial f} l$



Proposition 1. *∂f is the smallest convex valued upper semi-continuous function containing df whenever it is defined.*

Proof. First, ∂f is upper semi-continuous : by compactness, is suffices to show that for all $x \in U, \varepsilon > 0$, for y sufficiently close to x.

$$\partial_y f \subset \partial_x f + B(0, \varepsilon)$$

Assume for contradiction that there exists a sequence $x_i \to x$ such that for all i, $\partial_{x_i} f$ has an element out of $A := \partial_x f + \overline{B}(0, \varepsilon/2)$. A being convex and closed, let x_i' be such that $d_{x_i'} f$ is out of A, and such that $x_i' - x_i \to 0$. Up to extraction, $d_{x_i'} f$ converges to an element in $\partial_x f$, so it enters A, contradiction.
The second part follow from ∂f being a convex hull. $\square$

The subdifferential allows us to generalize the idea of regular and critical points, which will be an important tool for generalizing Morse theory. The definition is as follows :

Definition 3. *A point $x \in U$ will be called critical if there exists some $l \in \partial_x f$ not of maximal rank. x will be called regular if it isn't critical.*

To extend these subdifferentiability notions to manifolds, one must guarantee that for all charts, the definitions of ∂f coincide. In this is the case, $\partial_x f$ is a compact convex subset of $T_x^* M$. The following generalization of the chain rule allows us to do that.

Proposition 2 (Subdifferential chain rule)**.** *Let $h : \mathbb{R}^n \to \mathbb{R}^k$ be a C^1 function. For $x \in U$:*

$$\partial_x(h \circ f) \subset d_{f(x)}h \circ \partial_x f$$

If h is a diffeomorpism, we have equality.

Proof. For the first part, because f is differentiable a.e, it suffices to show that the leftwise term is an upper semi-continuous convex valued function that contains $d(h \circ f)$ whenever df is defined. This follows from the definition of ∂f.

If h is a diffeomorphism, then for $x \in U$:

$$\partial_x f = \partial_x(h^{-1} \circ h \circ f) \subset d_{h \circ f(x)}h^{-1} \circ \partial_x(h \circ f) \subset \partial_x f$$

so we have equalities, and the result follows. □

Although some important differentiablity related results can be obtained with locally Lipschitz functions as we'll see, a general locally Lipschitz function can have some pathological behaviours. As an example, it's subdifferential isn't in general a topological submanifold of M : define

$$A = \bigcup_{n \geq 0} \left[\frac{1}{2^{2n+1}}, \frac{1}{2^{2n}} \right[, B = \bigcup_{n \geq 0} \left[\frac{1}{2^{2n+2}}, \frac{1}{2^{2n+1}} \right[$$

The integral of the following function constitutes a counterexample, no chart can be found around the point $(0, 1)$ which is a point in the subdifferential.

$$\begin{array}{rcl} \mathbb{R} & \longrightarrow & \mathbb{R} \\ x & \longmapsto & \begin{cases} 2 & x \in A \\ 1 & x \in B \\ 0 & x \leq 0 \text{ or } x \geq 1 \end{cases} \end{array}$$

However, the functions we'll study, for example continuous selections (see Section 2) have much nicer behaviour and we retrieve this property.

1.1.2 The case of convex functions

If we consider convex Lipschitz functions $f : U \subset \mathbb{R}^m \to \mathbb{R}$, where U is a convex open set, the subdifferential can be understood much more precisely (in fact, this object was first invented for convex functions).

Consider a point $x \in U$ where $d_x f$ exists. By convexity, we have

$$\forall z \in U, f(z) \geq f(x) + d_x f(z - x)$$

Denote C_x the set of all linear maps verifying such property at a point x. The function $C : x \mapsto C_x$ is convex-valued, upper semi-continuous and contains d_x whenever it is defined. Moreover, at such points we have for h small enough :

$$f(x + h) = f(x) + d_x f(h) + o(h) \geq f(x) + \varphi(h)$$

where $\varphi \in C_x$. We deduce $(\varphi - d_x f)(h) = o(h)$, so $\varphi = d_x f$. Upper semi-continuity gives us $C \subset \partial f$, and the reciprocal is given by minimality of ∂f. So for a convex function the subdifferential can be described as

$$\partial_x f = \{\varphi \in (\mathbb{R}^m)^* \mid \forall z \in U, f(z) \geq f(x) + \varphi(z - x)\}$$

which gives a geometrical meaning to the name subdifferential.

Lastly, we connect the subdifferential to the directionnal derivatives of f. The directionnal derivative of f at x in direction $v \in \mathbb{R}^m$ is given by

$$D_x f(v) = \lim_{t \to 0^+} \frac{f(x + tv) - f(x)}{t}$$

and is always defined since f is convex. The two concepts are related by the following formula

$$D_x f(v) = \max_{l \in \partial f} \langle l, v \rangle$$

1.2 Generalized differential theorems

We now consider the specific case of $f : U \subset \mathbb{R}^m \to \mathbb{R}^m$. This section is devoted to proving generalizations by Clarke [6] of the inverse function theorem and the implicit function theorem for the case of Lipschitz functions.

In this section, we norm the space of linear maps $\mathbb{R}^m \to \mathbb{R}^m$, which we denote $\mathcal{L}(\mathbb{R}^n)$, with the operator norm $\|\cdot\|$ obtained from the norm used on $\mathbb{R}^n$.

Lemma 1. *Assume U is convex, and that there exists some closed convex set C such that*

- *Every $l \in C$ is invertible.*
- *$\partial_x f \subset C$ for all $x \in U$*

Then there exists some $\delta > 0$ such that for all $x, y \in U$:

$$|f(x) - f(y)| \geq \delta |x - y|$$

Proof. Firstly, we set

$$\delta = \inf \{|l(h)| \mid l \in C, h \in S\}$$

where S is the unit sphere in $\mathbb{R}^n$. $\delta > 0$ because all $l \in C$ are invertible and $C \times S$ is closed.

Let $x, y \in U$ and assume $x \neq y$. Let π be the plane perpendicular to $y - x$ passing through x. f being differentiable a.e, Fubini's theorem implies that for almost all $x' \in \pi$, f is differentiable almost on the whole line passing through x' and perpendicular to π (i.e f isn't differentiable on a subset of this line of one dimensional measure 0).
Denoting $F : t \longmapsto f(x' + t(y - x))$, we have :

$$f(x' + y - x) - f(x') = \int_0^1 F'(t)dt = \int_0^1 d_{x'+t(y-x)} f(y - x)dt \tag{1}$$

Define the linear map

$$L = \int_0^1 d_{x'+t(y-x)} f dt$$

C being convex and closed, $L \in C$. Observe that the integral on the right of (1) is just $L(y - x)$. So

$$|f(x' + y - x) - f(x')| \geq \delta |y - x|$$

This inequality holds for almost all $x' \in \pi$, so by continuity of f, it holds for x and we get the desired result. □

Theorem 1 (Lipschitz inverse function theorem)**.** *If x_0 is a regular point of f, then there exists V, W, neighbourhoods of x_0 and $f(x_0)$ respectively, such that f induces a bijection $f : V \to W$ where f^{-1} is Lipschitz too.*

Proof. By compactness of $\partial_{x_0} f$, regularity of x_0, and the fact that the invertible linear maps form an open set, there exist some $\varepsilon > 0$ such that the set

$$C := \partial_{x_0} f + \overline{B}(0, \varepsilon)$$

is composed only of invertible maps. By upper semi-continuity of ∂f, there is some $r > 0$ such that for $y \in B(x_0, r)$, $\partial f(y) \subset C$. We then apply Lemma 1 with $U = B(x_0, r)$ and get a corresponding $\delta > 0$.

Set $V = B(x_0, r/2)$ and $W = f(V)$. We showed in particular that f is injective on $\overline{V}$, by compactness $f : \overline{V} \to f(\overline{V})$ induces a homeomorphism. W is then open. The inequality we got with δ implies that f^{-1} is $\frac{1}{\delta}$ Lipschitz. □

From this theorem, follows the implicit function theorem for Lipschitz functions

Theorem 2. *Let $F : U \to \mathbb{R}^n$, where $U \subset \mathbb{R}^m \times \mathbb{R}^n$ is open and F is Lipschitz. Let $(x_0, y_0) \in U$ such that $F(x_0, y_0) = 0$ and y_0 is a regular point of $y \mapsto F(x_0, y)$. Then there exist neighbourhoods $(x_0, y_0) \in W \subset U$, $x_0 \in V \subset R^m$ and a Lipschitz function $g : V \to \mathbb{R}^n$ such that*

$$(x, y) \in W \text{ and } F(x, y) = 0 \Leftrightarrow x \in V \text{ and } y = g(x)$$

The proof is almost the same as the usual implicit function theorem, where the continuity of the differential is replaced by the semi continuity of ∂F.

1.3 First Morse lemma for locally Lipschitz functions

The goal of this section is to use the generalized notion of critical points in order to generalize the first Morse lemma to the case of locally Lipschitz functions. Fix M an m-dimensional smooth manifold, $f : M \to \mathbb{R}$ a continuous function and define for all $c \in \mathbb{R}$ the sublevel set of f of value c as

$$M_c := f^{-1}(]-\infty, c])$$

The classical first Morse lemma states that if f is C^1, and there are no critical points in $f^{-1}([a, b])$ (that we assume to be compact), then M_b is diffeomorphic to M_a and M_a is a deformation retract of M_b. We extend this to locally Lipschitz functions, following [1]. Note that in this setting x is a critical point of f iff $0 \in \partial_x f \subset T_x^* M$.

Proposition 3. *Let $a < b \in \mathbb{R}$ such that $f^{-1}([a, b])$ is compact and free of critical points. Then there exists a locally Lipschitz homotopy $H : M_b \times [0, 1] \to M_b$ such that for all $s \in [0, 1]$, $H(\cdot, s)$ is a homeomorphism verifying*

$$H(M_b \times \{s\}) \subset M_{b(1-s)+as}$$

Proof. We'll construct $V \in \mathrm{Vec}(M)$ a smooth complete vector field on M such that, for all $x \in M$ such that $d_x f$ is defined :

$$d_x f(V(x)) \leq 0 \text{ and } d_x f(V(x)) \leq -1 \text{ if } x \in f^{-1}([a, b]) \tag{2}$$

Let $x \in f^{-1}([a, b])$. x being regular and $\partial_x f$ convex, we find an affine hyperplane in $T_x^* M$ separating it from 0. This gives us $v_x \in T_x M$ such that $\langle \partial_x f, v_x \rangle \leq -2$. By semi continuity, there exists an open neighbourhood U_x of x such that for all $y \in U_x$, $\langle \partial_y f, v_x \rangle \leq -1$. We choose this neighbourhood such that it has compact closure.

These U_x cover $f^{-1}([a, b])$. Let $U_{x_1}, ..., U_{x_r}$ be a finite subcover, and let $\rho_1, ..., \rho_r$ be a partition of unity subordinated to this covering. We set

$$\forall x \in M, V(x) = \sum_{i=1}^{r} \rho_i(x) v_{x_i}$$

which indeed verifies (2) : if $d_x f$ is defined, it is contained in $\partial_x f$, and V is of compact support so it is complete.

Now observe that for $x \in f^{-1}([a,b])$, $t \mapsto f(e^{tV}x)$ is strictly decreasing. Denote by $t(x,s)$, $s \in [0,1]$, the unique time such that $f(e^{t(x,s)V}x) = as + f(x)(1-s)$. The equation defining $t(x,s)$ is locally Lipschitz in s, x and t, so by the Lipschitz implicit function theorem, t is a locally Lipschitz function of (x,s).

We can similarly define t' a locally Lipschitz function of x such that $f(e^{t'(x)V}x) = b$ for x in the orbit of some $y \in f^{-1}(b)$ by V. Denote O the set of $x \in M$ that are in such orbits. The homotopy is then given by

$$H : \begin{array}{ccc} M_b \times [0,1] & \longrightarrow & M_b \\ (x,s) & \longmapsto & \begin{cases} e^{t(e^{t'(x)V}x,s)V}x & \text{if } x \in O \\ x & \text{else} \end{cases} \end{array}$$

$\square$

2 Morse theory of continuous selections

2.1 Definitions

We now restrict ourselves to continuous functions that are piecewise smooth with finitely many pieces. We define those as continuous selections.

Definition 4. *A continuous function $f : M \to \mathbb{R}$ is a continuous selection if there exists $\varphi_1, ..., \varphi_n$ "regular" functions such that*

$$\forall x \in M, f(x) \in \{\varphi_1(x), ..., \varphi_n(x)\}$$

The term "regular" is used loosely here to indicate that the φ_i satisfy some extra condition on top of being continuous. For what follows, we'll restrict ourselves to continuous selections of smooth functions (but C^2 will suffice).

Note that a continous selection of locally Lipschitz functions is itself locally Lipschitz : if M is an open set of $\mathbb{R}^m$, and we assume that the functions φ_i are Lipschitz with constants c_i, then f is Lipschitz with constant $\max_i c_i$. Because smooth functions are locally Lipschitz, the functions we'll study will be too.

We can then relate ∂f to the differentials of the φ_i. Indeed, a limit of points of the form $d_x\varphi_i$ in T^*M is of the same form. Define the set of active indices at x,

$$I(x) = \{i \in [\![1, n]\!] \mid \varphi_i(x) = f(x)\}$$

$\partial_x f$ is then the convex hull of some $d_x\varphi_i$, $i \in I(x)$. It is possible that not all $d_x\varphi_i$ are used, if for example one of the functions isn't active in a neighbourhood of x.

We would like to define what it means for such a continuous selection to be Morse. Let f be a continuous selection of $\varphi_1, ..., \varphi_n$. Following [1], we give three conditions for a function to be considered Morse.

$$\text{The graphs of the } \varphi_i, i \in [\![1, n]\!], \text{ are transversal} \tag{C1}$$

This condition has some important consequences : for a non empty $I \subset [\![1, n]\!]$, we define $W_I := \{x \in M \mid I(x) = I\}$. Condition (C1) implies that W_I is a submanifold of M of codimension $|I| - 1$: the intersection of the graphs of $\varphi_i, i \in I$ is a submanifold of $M \times \mathbb{R}$ of codimension $|I|$ and W_I is the pre-image of this intersection through the diffeomorphism $x \mapsto (x, \varphi_i(x))$, $i \in I$, with the codomain being the graph of φ_i.

This transversality condition also implies that for $x \in M$, the differentials $d_x\varphi_i$ are affinely independant where $i \in I(x)$. By affinely independant, we mean that one of the two equivalent conditions are satisfied : for some/any $j \in I(x)$, the family $(d_x\varphi_i - d_x\varphi_j)_{\substack{i \in I(x) \\ j \neq i}}$ is linearly independant, or equivalently, the family

$$\begin{pmatrix} d_x\varphi_i \\ 1 \end{pmatrix}, i \in I(x)$$

is linearly independant. This implies that $\partial_x f$ is a simplex with less than $|I(x)| - 1$ vertices.

Note that for all $I \subset [\![1, n]\!]$, $f_{|W_I}$ is a smooth function. This motivates the second condition

$$\text{For all non empty } I \subset [\![1, n]\!], f_{|W_I} \text{ is Morse} \tag{C2}$$

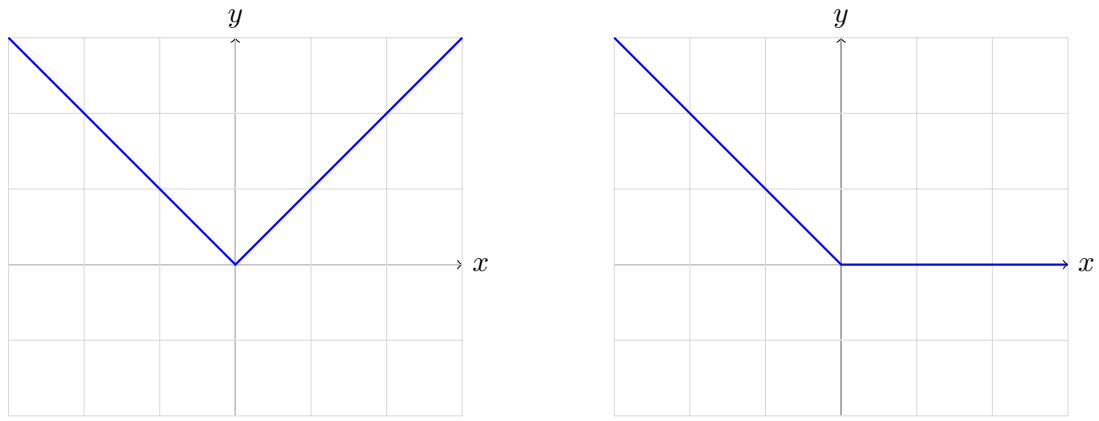
Consider $x \in M$ a critical point of f. Observe that x must also be a critical point of $f_{|W_{I(x)}}$. So $T_x W_{I(x)} \subset \bigcap_{i \in I(x)} d_x\varphi_i$, but the converse can fail if the $d_x\varphi_i$ are not linearly independant enough. The last condition fixes that

$$\forall x \in M, \forall j \in I(x), (d_x\varphi_i)_{\substack{i \in I(x) \\ i \neq j}} \text{ are linearly independant} \tag{C3}$$

Geometrically, this condition corresponds to requiring that 0 is either outside or in the interior of the simplex $\mathrm{conv}\{d_x\varphi_i \mid i \in I(x)\}$, but not on one of its faces. Note that this condition can be relaxed by restricting it to the subset of $I(x)$ participating in $\partial_x f$.

A continuous selection f is then called a Morse function if it verifies all three conditions. Note that as in the smooth case, a generic continuous selection is Morse : (C1) is a generic condition, any two functions can be slightly perturbed in order for their graphs to be transversal, and any small perturbation of transversal graphs preserves transversality. By genericity of smooth Morse functions, (C2) is also a generic condition. With the geometric interpretation, we can also see that condition (C3) is generic.

Figure 2: Example and counterexample of condition (C3)



2.2 Normal form of a Morse continuous selection

We show here an equivalent of the second Morse lemma for the case of Morse continuous selections. Noticing that $x \mapsto I(x)$ is upper semi-continuous and using the definitions yields the following result.

Lemma 2. *If $f : M \to \mathbb{R}$ is a Morse continuous selection, all its critical points are isolated.*

We'll then consider, up to restricting ourselves to a small enough chart, that $f : \mathbb{R}^m \to \mathbb{R}$ is Morse and has no critical points other than 0. We'll write for shorthand $W := W_{I(0)}$, $k := |I(0)| - 1$, $c := f(0)$, and assume that $I(0) = [\![1, k+1]\!]$.

Theorem 3. *If 0 is a critical point of f, then there exists Lipschitz local coordinates around 0, $\psi : \mathbb{R}^k \times \mathbb{R}^{m-k} \to U$ such that*

$$\forall y \in \mathbb{R}^k, \forall z \in \mathbb{R}^{m-k}, f(\psi(y,z)) = \lambda(y) + q(z)$$

where λ is a continuous selection of $k+1$ affinely independant linear functions, and q is of the form

$$\forall z \in \mathbb{R}^{m-k}, q(z) = c - \sum_{i=1}^{\mu} z_i^2 + \sum_{i=\mu+1}^{m-k} z_i^2$$

with $0 \leq \mu \leq m-k$.

Proof. We first restrict ourselves to O a neighbourhood of 0 such that W can be considered rectified in some smooth coordinates i.e, $W \cap O = 0 \times \mathbb{R}^{m-k}$. We can also use the smooth Morse lemma to get $f(0,z) = q(z)$. For $(y,z) \in \mathbb{R}^m$, write $f^0(y,z) = f(y,z) - f(0,z)$. We will build for all z close enough to 0 a locally invertible map $\psi_z : \mathbb{R}^k \to \mathbb{R}^k$ such that $\psi_z(0) = 0$ and for y close enough to 0, $f^0(\psi_z(y), z)$ is a continuous selection of linear maps independant of z.

For what follows we'll fix z and omit it in the notations for convenience. We find a neighbourhood U of $0 \in \mathbb{R}^k$ such that

- U is connected.
- For all $y \in U$, $I(y) \subset [\![1, k+1]\!]$.
- For all $y \in U$, any k $d_y\varphi_i$ are linearly independant
- There exists diffeomorphisms $\psi_j : U \to V_j$ such that for all $i \neq j$, $\psi_j(\varphi_i(y)) = \varphi_i(0) + d_0\varphi_i(y)$, $y \in U$.

We now want to combine these to take account of the possible intersections between the O_j. Our goal is to combine to ψ_j together with smooth coefficients $\mu_j : U\backslash\{0\} \to \mathbb{R}$ of sum 1 and set our final diffeomorphism to be

$$\psi(y) = \sum_{i=1}^{k+1} \mu_i(y)\psi_i(y)$$

Moreover, ψ should take all active φ_i to its affine approximation.
Let $\Omega \subset U$ be a maximal set on which we can define such functions, given by Zorn's lemma. Assume there exists some non zero $y \in \Omega$ verifying that all of these exist at y. Let j be such that $y \in O_j$

$$F : \begin{array}{ccc} U \times \mathbb{R}^{k+1}_{<2} & \longrightarrow & \mathbb{R}^{k+1} \\ (y, \mu_1, ..., \mu_{k+1}) & \longmapsto & \left(\left(\varphi_i \left(\sum_{r=1}^{k+1} \mu_r \psi_r(y) \right) - \varphi_i(0) - d_0\varphi_i(y) \right)_{i \neq j}, \sum_{i=1}^{k+1} \mu_i - 1 \right) \end{array}$$

We'll assume for convenience for what follows that the $\psi_i(y)$ are affinely independant. A modified version of the argument also works if the ψ_i are allowed to be affinely dependant, where one defines μ on the closed subset of $U\backslash\{0\}$ where the ψ_i aren't affinely independant by setting all of them except one to be 0. This can be achieved in a smooth way with smooth interpolators that nullify all μ except one in this set.

By assumption we have $F(y, \mu_1(y), ..., \mu_{k+1}(y)) = 0$. Then by condition C3, the differential of F along μ is invertible. So by the implicit function theorem, there exists a neighbourhood of y on which all μ_i can be extended. So Ω is open in $U\backslash\{0\}$.

Ω is also closed in this set : assume there is a sequence (y_r) in Ω converging to some non zero $y \in O_j$ for some j. By construction, the μ_i are bounded, so we can assume up to extraction that $\mu(y_r) \to \tilde{\mu}$. Then $F(y, \tilde{\mu_1}, ..., \tilde{\mu_{k+1}}) = 0$, and we again use the implicit function theorem to extend the μ_i to y. This means $y \in \Omega$.

Because $U\backslash\{0\}$ is connected, $\Omega = U\backslash\{0\}$ because we can trivially define μ_i at a point where only one index isn't active. ψ is then a local diffeomorphism using the fact that $d_0\varphi_i \circ d_0\psi = d_0\varphi_i$ for all i.

We constructed for all z close enough to 0 to approximate $d_{(0,z)}\varphi_i$ by $d_0\varphi_i$, ψ_z. They verify the following locally Lipschitz equation : $f(\psi_z(y), z) = \lambda(y) + q(z)$ where λ is a continuous selection of $d_0\varphi_i$. So $(z, y) \mapsto \psi_z(y)$ is locally Lipschitz.

□

We now try to understand better the structure of the non smooth component that appears in the normal form theorem. It turns out that it can be encoded in a purely combinatorial way, as the following theorem shows [8].

Theorem 4. *Let $\lambda : \mathbb{R}^k \to \mathbb{R}$ be a continuous selection of $k+1$ affinely independant linear functions, $\lambda_1, ..., \lambda_{k+1}$. Then there exists a unique family $S_1, ..., S_l$ of subsets of $[\![1, k+1]\!]$ verifying that*

- *$S_i \not\subset S_j$, $i \neq j$*
- *λ can be written as*

$$\lambda(y) = \min_i \max_{j \in S_i} \lambda_j(y)$$

Proof. Let's first tackle uniqueness. Let $S_1, ..., S_l$, $D_1, ..., D_r$ be two such distinct families. Assume without loss of generality that $S_1 \neq D_j$. We can also assume that for all $1 \leq j \leq r$, $S_1 \not\supset D_j$. If it is the case, then for all $1 < i \leq l$ we have $D_j \not\supset S_i$ by assumption on S. We can then swap S and D to get what we want.
By affine independance, the following linear system has a solution

$$\lambda_i(y) + \mu = c_i, i \in [\![1, k+1]\!]$$

where the unknowns are y, μ and $c_i = 0$ if $i \in S_1$ and 1 otherwise. It follows that, if we use the representation with D_i, we get $\lambda(y) = 1 - \mu$ and if we use the one with the S_i, $\lambda(y) = -\mu$.

Now for existence. For all permutation $\nu \in \mathfrak{S}_{k+1}$ define the the cone

$$K_\nu = \{y \in \mathbb{R}^k \mid \lambda_{\nu(1)}(y) \leq ... \leq \lambda_{\nu(k+1)}(y)\}$$

and the function $s : \nu \in \mathfrak{S}_{k+1} \mapsto i \in [\![1, k+1]\!]$ where i is such that $\lambda_{\nu(i)} = \lambda$ on K_ν. If we assume that for all y and ν we have

$$\lambda(y) \leq \max\{\lambda_{\nu(1)}(y), ..., \lambda_{\nu(s(\nu))}(y)\}$$

then it suffices to consider the family of set $S_\nu = \{\nu(1), ..., \nu(s(\nu))\}$ and simplify these by removing those who are included in one another.

We now give a broad idea of the proof of the claim. The result is true by definition on K_ν. For any $y \in \mathbb{R}^k$, set $y' \in K_\nu$ and consider the evolution of the order of λ_i along the line between y and y' : the value of each λ_i evolves at a constant rate. Whenever the current λ_i defining λ encounters another λ_j and λ becomes defined as λ_j afterwards, either λ_j is smaller than λ_i after the encounter, or the opposite. If we denote I the set of indices such that the functions are smaller than λ_j before the encounter, then in either case we see that λ stays smaller than the maximum of $\lambda_{i'}$ $i' \in I$. This concludes the proof because the line starts at K_ν. □

2.3 Describing the homology of sublevel sets

Our goal for this section is to describe more precisely the transition that happens as the critical point of f is passed, where f is still a Morse continuous selection defined on $\mathbb{R}^m$ with 0 as its only critical point.

The normal form of f and theorem 4 allows us to fully describe the behaviour of f with purely combinatorial information, a smooth Morse index and a more complicated set representation. We can encode the combinatorial structure that appears in its non smooth part in a way that will give us topological and homotopical informations about the sublevel sets : define the simplical complex $K(S_1^c, ..., S_l^c)$ on $k+1$ vertices whose faces are given by the complements in $[\![1, k+1]\!]$ of the sets S_i describing the non smooth part of f.

Another thing to note is that when put in normal form, f is a quasi homogeneous function, i.e there exists some real numbers $\alpha_1, ..., \alpha_m > 0$ such that for all $t > 0, x \in \mathbb{R}^m$, we have

$$f(t^{\alpha_1} x_1, ..., t^{\alpha_m} x_m) = tf(x)$$

In this case we have $f(ty, t^{1/2}z) = tf(y, z)$ if we assume that $f(0) = 0$, assumption we'll make without loss of generality. For a quasi homogenous function h, we'll denote δ_t the power-scaling map such that $h(\delta_t x) = th(x)$. The following lemmas on quasi homogeneous functions will prove useful for describing the homotopy type of sublevel sets.

Lemma 3. *Let $h : \mathbb{R}^m \to \mathbb{R}$ be a quasi homogeneous locally Lipschitz function having only 0 as a critical point. We have the following homotopy equivalence*

$$(M_1, M_{-1}) \simeq (C_{h^{-1}(-1)}, h^{-1}(-1))$$

where M_1, M_{-1} are the sublevel sets and $C_{h^{-1}(-1)}$ is the cone over $h^{-1}(-1)$.

Proof. Let $\mathbb{S}$ denote the unit sphere. Define a smooth V over $\mathbb{S}$ such that for all $x \in \mathbb{S}$, $V(x) \in \mathbb{R}^m$ and $\langle \partial_x h, V(x) \rangle \leq -1$. Extend V into a vector field on $\mathbb{R}$ such that it commutes with δ_t for all $t > 0$, i.e for $x \in \mathbb{S}$, $V(\delta_t x) = d_x \delta_t(V(x))$. It follows that $s \mapsto h(e^{sV}x)$ is strictly decreasing for all $x \neq 0$.

From this observation, we define similarly to lemma 3, for all $x \in h^{-1}(]-\infty, 0[)$, $p(x)$, the only point of the orbit of x in $h^{-1}(-1)$. We also define for all $x \in h^{-1}(-1), s \in [0, 1[$, $t(x, s)$ such that it verifies $h(e^{-t(x,s)V}x) = h(x) + s$. If we then denote by $\pi : [0, 1] \times h^{-1}(-1) \to C_{h^{-1}(-1)}$ the projection map, where the top of the cone is the point at height $s = 1$, the homotopy equivalence maps are the following

$$F : \begin{array}{ccc} M_1 & \longrightarrow & C_{h^{-1}(-1)} \\ x & \longmapsto & \begin{cases} \pi(0, p(x)) & x \in M_{-1} \\ \pi(h(x) + 1, p(x)) & x \in h^{-1}([-1, 0]) \\ \pi(1, 0) & x \in h^{-1}([0, 1]) \end{cases} \end{array}$$

$$G : \begin{array}{ccc} C_{h^{-1}(-1)} & \longrightarrow & M_1 \\ \pi(s, x) & \longmapsto & e^{-t(x,s)}x \end{array}$$

We can see that $F \circ G = \mathrm{id}_{C_{h^{-1}(-1)}}$. $G \circ F$ is homotopic to the identity : we can use V to contract M_{-1} onto $h^{-1}(-1)$, and also to retract $h^{-1}([0, 1])$ onto $h^{-1}(0)$. □

If X and Y are two topological spaces, we define the join of X and Y, denoted by $X * Y$, as the quotient space of $X \times Y \times [0, 1]$ by the following equivalence relation

$$\forall x, x' \in X, \forall y, y' \in Y, (x, y, 0) \sim (x, y', 0) \text{ and } (x, y, 1) \sim (x', y, 1)$$

Intuitively, we join together the spaces X and Y with all possible segments between the two. Some examples of join include $X * \{y\}$ wich is the cone of X, $X * \mathbb{S}^0$ which is the suspension of X. More generally the join of X with $\mathbb{S}^n$ is an $n + 1$-fold suspension.

By using the same ideas as in lemma 3, one can prove the following useful lemma.

Lemma 4. *Let $h : \mathbb{R}^k \times \mathbb{R}^{m-k} \to \mathbb{R}$ be a quasi homogeneous function such that for all $(y, z) \in \mathbb{R}^m$, we have $h(y, z) = a(y) + b(z)$ where a and b are quasi-homogeneous locally Lipschitz with respective exponents $\alpha, \beta > 0$. We have the following homotopy equivalence*

$$h^{-1}(-1) \simeq a^{-1}(-1) * b^{-1}(-1)$$

If we write $f(y, z) = \lambda(y) + q(z)$ as described by the normal form theorem, what remains to be done to understand the relative homology of (M_1, M_{-1}) is to describe $q^{-1}(-1)$ and $\lambda^{-1}(-1)$. The first one has the homotopy type of $\mathbb{S}^\mu$ where μ is the smooth Morse index of f, by classical Morse theory.

For the second, note that we have the following homotopy equivalence $\lambda^{-1}(-1) \simeq \lambda^{-1}(]-\infty, 0[)$. Using the min max form, we get

$$\lambda^{-1}(]-\infty, 0[) = \bigcup_i \bigcap_{j \in S_i} \lambda_j(]-\infty, 0[)$$

Note that each intersection is a convex cone we'll denote by C_i. We'll relate this to the simplical complex $K(S_1^c, ..., S_l^c)$ by using the nerve of the family $C = (C_i)_i$, i.e we'll look at the following set

$$N(C) := \{J \subset [\![1, l]\!] \mid \bigcap_{j \in J} C_j \neq \varnothing\}$$

From the next lemma, it follows that the nerve of $N(C)$ is exactly the one of the family of sets $S_1^c, ..., S_l^c$.

Lemma 5. *Let $J \subset [\![1, k + 1]\!]$. J is the whole set if and only if*

$$\bigcap_{j \in J} \lambda_j^{-1}(]-\infty, 0[) = \varnothing$$

Proof. We first show that if $J = [\![1, k+1]\!]$, we indeed get an empty intersection. Recall that 0 is a critical point of f, so it is a critical point of λ. We then have

$$0 \in \partial_0 \lambda \subset \mathrm{conv}\{\lambda_1, ..., \lambda_{k+1}\}$$

So there exists $\mu_1, ..., \mu_{k+1}$ with sum 1 such that $0 = \sum_i \mu_i \lambda_i$. We then see that all λ_i cannot be strictly negative at the same time.

We now show that if $J \neq [\![1, k+1]\!]$, the intersection is not empty. Condition (C3) gives us linear independance of the family $(\lambda_j)_{j \in J}$. The linear system $\lambda_j(y) = -1, j \in J$ then has a solution, where the unknown is y. □

Because every C_i is convex, their intersections are contractible. Applying Leray's nerve theorem (cite), $\lambda^{-1}(-1)$ has the homotopy type of the simplical complex given by the nerve. By virtue of what we discussed, this complex is exactly $K(S_1^c, ..., S_l^c)$. The homotopy type of $f^{-1}(-1)$ is then given by

$$f^{-1}(-1) \simeq K(S_1^c, ..., S_l^c) * \mathbb{S}^\mu$$

By recalling that a join with a sphere is a suspension, we get the following result in homology.

Theorem 5. *For $i \geq 0$, we have*

$$H_{i+\mu}(M_1, M_{-1}) = \tilde{H}_{i-1}(K(S_1^c, ..., S_k^c))$$

3 Euclidian distance from definable sets

In this section, we will work in the euclidian space $\mathbb{R}^m$. Some of what we'll do here can be generalized to a more general Riemannian manifold. All throughout, $X \subset \mathbb{R}^m$ will be a smooth manifold unless specified otherwise, $Y \subset \mathbb{R}^m$ a closed, definable (whatever we decide it to mean) set. The main function we will study is the euclidian distance from Y restricted to X, i.e

$$\forall x \in X, \mathrm{dist}_Y(x) = \min_{y \in Y} |x - y|$$

where $|\cdot|$ is the euclidian norm. This function is locally Lipschitz. In what will follow, we identify $(\mathbb{R}^m)^*$ with $\mathbb{R}^m$ via the euclidian structure. $\partial_x(\mathrm{dist}_{Y|X})$ will often be identified with a subset of T_xX, where $x \in X$.

Our main objectives will be to understand the subdifferential of this function and to find a setting where Morse theory of continuous selections can be applied. We will also study more closely the particular case where Y is a point and X is algebraic.

3.1 Differentiability and subdifferential of the distance function

3.1.1 Computing the subdifferential in euclidian space

We first study the easier case where $X = \mathbb{R}^m \backslash Y$, which gives rise to a very nice and intuitive geometric picture. We first require some results on "infinite continuous selections", which is indeed how dist_Y is defined.

Let $\varphi : \mathbb{R}^m \times Z \to \mathbb{R}$ be a continuous function, where Z is a compact space. We can define a family $(\varphi_z)_{z \in Z}$ of continuous functions, where $\varphi_z = \varphi(\cdot, z)$. Similar to the finite case we can define a "compact continuous selection" as a continuous f such that for all $x \in \mathbb{R}^m$

$$f(x) \in \{\varphi_z(x) \mid z \in Z\}$$

Similarly to the finite case, we define the set of active indices at x, $Z(x)$ as the set of all z that take the value of f at x. We have the following result for a special kind of these functions.

Theorem 6 (Danskin's [3])**.** *Assume that for all z, φ_z is differentiable, $z \mapsto d_x\varphi_z$ is continuous and $f = \max_{z \in Z} \varphi_z$.*

1. *f is convex and for all $x, v \in \mathbb{R}^m$*
$$D_x f(v) = \max_{z \in Z(x)} d_x\varphi_z(v)$$

2. *f is locally Lipschitz and we have*
$$\partial_x f = \mathrm{conv}\{d_x\varphi_z(v) \mid z \in Z(x)\}$$

Proof. We'll need 1. for 2.

1. The convexity of f is a classical result from convex analysis. Notice that for $t > 0$
$$\frac{f(x+tv) - f(x)}{t} \geq \frac{\varphi_z(x+tv) - \varphi_z(x)}{t}$$
for $z \in Z(x)$, so we get $D_x f(v) \geq d_x\varphi_z(v)$. For the other inequality, consider a positive sequence $t_n \to 0$. For all n, select $z_n \in Z(x + t_n v)$, by compactness assume it converges to $z^* \in Z$. By

maximality of f, we have $\varphi_{z^*}(x) \geq \varphi_z(x), z \in Z$, so $z^* \in Z(x)$. By convexity of f and φ_{z_n}, we get

$$\begin{aligned} D_x f(v) &\leq \frac{f(x+t_n v) - f(x)}{t_n} \\ &= \frac{\varphi_{z_n}(x+t_n v) - \varphi_z^*(x)}{t_n} \\ &\leq \frac{\varphi_{z_n}(x+t_n v) - \varphi_{z_n}((x+t_n v) - t_n v)}{t_n} \\ &\leq d_x \varphi_{z_n}(v) \\ &\to d_x \varphi_{z^*}(v) \end{aligned}$$

so $D_x f(v)$ is indeed the maximum.

2. Let $x \in \mathbb{R}^m$ and B be an open ball around x. Define for all $k > 0$ the following open set

$$V_k = \{z \in Z \mid \forall x, y \in \mathbb{R}^m, |\varphi_z(x) - \varphi_z(y)| < k|x-y|\}$$

B being of compact closure and all φ_z being C^1, they are all Lipschitz on B. Hence $\bigcup_{k>0} V_k = Z$, so by compactness, there is $k > 0$ such that $V_k = Z$. f is then Lipschitz on B with constant k.

We now show the equality involving ∂f. For that we use the results of the discussion in section 1.1.2. For the right to left inclusion, it suffices to observe that in virtue of 1., for $z \in Z(x)$

$$\forall v \in \mathbb{R}^m, f(x+v) \geq f(x) + d_x \varphi_z(v)$$

Assume that there is some $l \in \partial_x f$ not in the convex hull of $d_x \varphi_z$, $z \in Z(x)$. Because this convex hull is compact, we choose a strictly separating hyperplane : we choose v and c such that

$$\langle l, v \rangle > c > \langle d_x \varphi_z, v \rangle, z \in Z(x)$$

This contradicts the fact that $D_x f(v) = \max_{h \in \partial_x f} \langle h, v \rangle$, because $D_x f(v) = \max_{z \in Z(x)} \langle d_x \varphi_z, v \rangle$

□

We are finally ready to compute $\partial(\mathrm{dist}_Y)$. We will also denote by $Y(x)$ the set of all $y \in Y$ that are active at a point x.

Theorem 7. *Let $x \in \mathbb{R}^m \backslash Y$. We have*

$$\partial_x \mathrm{dist}_Y = \mathrm{conv}\left\{ \frac{x-y}{|x-y|} \mid y \in Y(x) \right\}$$

Proof. Let $\eta_Y = \frac{1}{2}\mathrm{dist}_Y^2$. We write for $x \in \mathbb{R}^m \backslash Y$

$$\eta_Y(x) = \frac{1}{2}|x|^2 - \left(\max_{y \in Y} \langle x, y \rangle - \frac{1}{2}|y|^2 \right)$$

The function $\varphi_y(x) = \langle x, y \rangle - \frac{1}{2}|y|^2$ is convex. If we denote by $h(x)$ the term in parenthesis, theorem 6 yields

$$\partial_x h = \mathrm{conv}\{\nabla_x \varphi_y \mid y \in Y(x)\} = \mathrm{conv}(Y(x))$$

We can apply the theorem because in a small neighbourhood of x, the active indices stay confined to a compact set.
By using that $\eta_Y = \frac{1}{2}|\cdot| - h$, we get

$$\partial_x \eta_Y = \{x - y \mid y \in Y(x)\}$$

If we now write $\mathrm{dist}_Y = \sqrt{2\eta_Y}$, and use the subdifferential chain rule applied to the diffeomorphism $\sqrt{\cdot}$ (because $\eta_Y(x) \neq 0$, $x \notin Y$) we get

$$\begin{aligned} \partial_x \mathrm{dist}_Y &= \frac{1}{\sqrt{2\eta_Y(x)}} \mathrm{conv}\{x - y \mid y \in Y(x)\} \\ &= \mathrm{conv}\left\{ \frac{x-y}{|x-y|} \mid y \in Y(x) \right\} \end{aligned}$$

because $\eta_Y(x) = \frac{1}{2}|y-x|^2$ for all $y \in Y(x)$ □

3.1.2 Special points and the general case of the subdifferential

We now consider $X \subset \mathbb{R}^m$ to be some non specific submanifold. Following [2], we define

Definition 5 (Medial axis)**.** *We define M_Y, the medial axis of Y as*

$$M_Y = \{x \in \mathbb{R}^m \mid |Y(x)| \geq 2\}$$

Some other special points of the distance function include the cut-locus of Y, that we'll denote $\mathrm{CLoc}(Y)$ and the central set of Y that we denote $C(Y)$, introduced respectively by Thom [7] and Milman cite.

Definition 6 (Central set)**.** *An open ball $B \subset \mathbb{R}^m \backslash Y$ will be called maximal if there is no open ball in $\mathbb{R}^m \backslash Y$ strictly containing B. The central set $C(Y)$ is then the set of the centers of all maximal balls.*

The stratification on Y (at least C^2), given by the fact it is definable, allows us to define the following

Definition 7 (Cut-locus)**.** *The cut locus of Y is defined as*

$$\mathrm{CLoc}(Y) = M_Y \cup \{x \in \mathbb{R}^n \mid y \in Y \mapsto |x-y|^2 \textit{ has a degenerate global minimum}\}$$

If we assume that Y is a submanifold of $\mathbb{R}^m$, we can link these seemingly differents concepts : in this case, $\overline{M_Y} = \mathrm{CLoc}(Y) = C(Y)$. $\overline{M_Y} = C(Y)$ can be found in [4]. One can prove $\overline{M_Y} \subset \mathrm{CLoc}(Y)$ by considering a sequence converging in $\overline{M_Y} \backslash M_Y$, and $\mathrm{CLoc}(Y) \subset C(Y)$ by considerations of oscullating circles for the case of degenerate points.

Returning to a more general definable set Y, the stratification on Y allows us to show that these equalities still hold, again by case inspection. The main argument is that in the degenerate/border case, everything happens relatively to one strata of Y.

These special sets of points play an important role in the differentiability of dist_Y as shown in the following proposition.

Proposition 4. *The function dist_Y is C^1 on $\mathbb{R}^m \backslash (Y \cup \overline{M_Y})$. Moreover, M_Y is definable and of codimension at least 1.*

Proof. Let $x \in \mathbb{R}^m \backslash (Y \cup \overline{M_Y})$. Let U be a neighbourhood of x contained in $Y \cup \overline{M_Y}$. On this neighbourhood, we can then define a function $\eta : U \to Y$ that to each $z \in U$ associates the unique $y \in Y(z)$. Because of that, we can safely assume that Y is itself a manifold, up to restricting ourselves to a strata. If we denote by $g : (a,b) \mapsto |a-b|^2$, and $g_a = g(a, \cdot)$, this y verifies the following C^1 equation.

$$\nabla_y (g_{z|Y}) = 0$$

Because z is not in $\mathrm{CLoc}(Y)$, differentiating this equality relatively to y yields an invertible matrix, so by the implicit function theorem η is C^1. It now suffices to write $\mathrm{dist}_Y(z) = \sqrt{g(\eta(z), z)}$ for $z \in U$.

Definablity is clear, the definition of M_Y can be written in first order logic by using the definability of Y. For the codimension, we use the convexity of the distance as follows : assume some $x \in M_Y$ admits a neighbourhood contained in such set. Fix $y_1 \neq y_2 \in Y(x)$. Let $z \neq x$ be a point in M_Y on the segment from x to y_2. Then, there must exist some $y_3 \in Y$ distinct from the other ones such that $y_3 \in Y(z)$. Then

$$|x - y_2| = |x - z| + |z - y_2| = \mathrm{dist}_Y(x) \leq |x - y_3|$$

Because $|z - y_2| = |z - y_3|$, we get by triangle inequality that $|x - y_3| = |x - y_2|$. By convexity, this must be the case on the whole segment from x to z. But this segment lies in the segment from x to y_2, so we must have $y_2 = y_3$, which is a contradiction. □

As a corollary, we can now compute the subdifferential of $\mathrm{dist}_{Y|X}$ for a general X, with some transversality constraints. If Y' is a definable set, we'll say that Y' is transversal to X if X is transversal to each strata of Y'.

Corollary 1. *If X is transversal to Y and $\overline{M_Y}$, then for all $x \in X$*

$$\partial_x(\mathrm{dist}_{Y|X}) = \mathrm{proj}_{T_xX}(\partial_x \mathrm{dist}_Y)$$

where proj_{T_xX} *is the orthogonal projection onto T_xX.*

Proof. Denote $g = \mathrm{dist}_Y$, and $f = \mathrm{dist}_{Y|X}$. We make the following observations :

- If g is differentiable at a point $x \in X$, so is f and $d_xf = \mathrm{proj}_{T_xX}(d_xg)$.
- f and g are differentiable of differential 0 in $X \cap \mathrm{int}(Y)$. Hence we can assume without loss of generality that $X \cap \mathrm{int}(Y) = \varnothing$. The transversality condition then gives us that $X \cap (Y \cup \overline{M_Y})$ is of codimension at least 1 in X by proposition 4.
- The set of all $x \in X$ such that g is differentiable at x is of measure 0 in X by proposition 4 and the last observation.

To compute $\partial_x f$, it then suffices to consider limits of sequences $d_{x_k}f$ where x_k is a point of differentiability of g. For such a sequence converging to some $x \in X$, we have, by continuity of proj.

$$\begin{aligned}\lim_k d_{x_k}f &= \lim_k \mathrm{proj}_{T_{x_k}X}(d_{x_k}g)\\ &= \mathrm{proj}_{T_xX}(\lim_k d_{x_k}g)\end{aligned}$$

proj_{T_xX} being linear, we get

$$\partial_x f = \mathrm{conv}\{\mathrm{proj}_{T_xX}(l) \mid l \in \partial_x g\} = \mathrm{proj}_{T_xX}(\partial_x g)$$

□

3.2 Distance functions as locally finite continuous selections

In this section, $Y \subset \mathbb{R}^m$ will be a smooth submanifold, as X. Our goal will be to show that under certain assumptions, the distance function is locally a continuous selection of finitely many C^2 functions, and so enters the Morse theory framework developped in section 2.

We'll denote by φ the Riemannian exponential map restricted to NY, i.e for $(y, v) \in NY$, $\varphi(y, v) = y + v$. The following proposition from [2] uses this map to find points where dist_Y is locally a continuous selection.

Proposition 5. *Assume Y is smooth and let $x \in \mathbb{R}^m$ be a regular value of φ. Then $Y(x) = \{y_1, ..., y_n\}$ is finite. Moreover there exists*

- *V a neighbourhood of x*
- *For all i, W_i a neighbourhood of y_i such that $\overline{W_i}$ are disjoint*

that verify

$$\forall z \in V, \mathrm{dist}_Y(z) = \min_i \mathrm{dist}_{Y \cap \overline{W_i}}(z)$$

and all $\mathrm{dist}_{Y \cap \overline{W_i}}$ *are smooth on V.*

3.3 Euclidian distance degree

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